

Vinh Ly

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Summary

Staff software engineer with **17 years of experience** building and running distributed systems at Google, YouTube, and Whatnot. I have spent most of that time on the infrastructure side: the platforms, pipelines, and tooling that keep large-scale services up and cheap to operate. Comfortable across **AWS, GCP, and Azure**, and experienced with Kubernetes, Terraform, Python, and Go. I have led teams and also done the work myself, which is why I still write code most days.

Core Skills

Languages: Python, Go, TypeScript, C++

Infrastructure as Code: Terraform, Pulumi, CloudFormation

Containers and Orchestration: Kubernetes, Helm, Docker, ArgoCD

Cloud: AWS (EC2, ECS, EKS, Lambda, S3, RDS), GCP, Azure

CI/CD and Observability: GitHub Actions, CircleCI, GitLab CI, Prometheus, Grafana, OpenSearch

Security and Networking: IAM/RBAC, secrets management, TLS/mTLS, VPC design

Work Experience

Staff Software Engineer | YouTube | Remote | 02/2025 - Present

- Working on backend systems for YouTube's content delivery infrastructure, handling ingestion, transcoding pipelines, and CDN integration at scale.
- Improving monitoring and alerting for video processing workloads, targeting better signal-to-noise in on-call dashboards and faster detection of pipeline regressions.

Staff Software Engineer | Whatnot | Hybrid | 05/2024 - 02/2025

- Joined Whatnot during a period of rapid user growth and took over infrastructure reliability for the live commerce platform. Migrated core services from ECS to **EKS** and rewrote Helm charts to support multi-region deployment.
- Reduced deployment-related incidents by roughly 40% after introducing a staged rollout process with automated smoke tests in **GitHub Actions**.
- Built out **Prometheus and Grafana** dashboards for the services the team owned, cutting mean time to detection on platform errors from over 10 minutes to under 3.

Staff Software Engineer / Manager | Google | Remote | 03/2015 - 04/2024

- Led a team of 8 engineers responsible for **Google-scale distributed systems** underpinning internal developer platforms. Managed performance reviews, hiring, and technical direction for 2 years before returning to IC work.
- Owned a multi-year migration of on-premises workloads to **GCP**, decommissioning several data center racks and moving traffic to cloud-native services. Hardware cost savings ran into eight figures annually.
- Introduced **Terraform** as the team's IaC standard, replacing a mix of ad-hoc scripts. Cut infrastructure provisioning time from days to under an hour for new service environments.
- Worked closely with **SREs and security engineers** to implement IAM/RBAC policies and secrets management with HashiCorp Vault, reducing blast radius of credential leaks across the fleet.
- Served as technical lead for a platform migration that consolidated four internal CI/CD systems into a single **Kubernetes-based pipeline**. Developer teams reported noticeably less friction getting changes to production.

Software Engineer | Google | Remote | 02/2011 - 02/2015

- Worked on YouTube's backend, primarily the video ingestion and transcoding pipeline. Optimized the worker queue to better prioritize urgent encoding jobs during peak traffic windows.
- Collaborated with the CDN team on cache invalidation logic. The changes cut redundant origin fetches by a measurable amount during high-traffic events like product launches.

Software Design Engineer II | Microsoft | Remote | 10/2007 - 12/2011

- Worked on Visual Studio's internal build and test infrastructure, supporting the team that shipped VS 2010. Built tooling in C++ and Python to track flaky tests across the nightly run fleet.
- The test tracking system the team shipped handled millions of test records per release cycle, running on a SQL Server backend.

Software Design Engineer Intern | Microsoft | Remote | 06/2006 - 08/2006

- Built an internal reporting tool for the test infrastructure team. The tool was picked up and used by three other teams in the organization before I finished my internship.

Education

BS Computer Science | University of Texas at Austin | 2003 - 2007